Radiology Case Reports 21 (2026) 4311–4317

Available online at www.sciencedirect.com

ScienceDirect

journal homepage: www.elsevier.com/locate/radcr

Case Report

Two cases of traumatic aortic isthmus pseudoaneurysm successfully managed with endovascular repair: Multidisciplinary critical care and anesthetic perspective ☆,☆☆

Mohammed Taloubi, MD[a,b,*], *Iman Zahrou, MD*[a,b], *Saley M.M. Rabilou, MD*[a,e], *Mohammed Chafi, MD*[a,b], *Lamya A. Sayad, MD*[b,d], *Ghizlane E. Aidouni, MD*[a,b,c,d], *Houssam Bkiyar, MD*[a,b,c,d]

[a] *Intensive Care Unit, Mohammed VI University Hospital Center, Oujda, Morocco*
[b] *Faculty of Medicine and Pharmacy, Mohammed First University, Oujda, Morocco*
[c] *Mohammed First University Oujda, FMP Oujda, LAMCESM, Oujda, Morocco*
[d] *Emergency Unit of the Specialized Hospital, Mohammed VI University Hospital Center, Oujda, Morocco*
[e] *Department of Radiology, University Hospital Mohammed 6, Oujda, Morocco*

ARTICLE INFO

Article history:
Received 5 March 2026
Revised 31 May 2026
Accepted 4 June 2026

Keywords:
Aortic isthmus pseudoaneurysm
Blunt thoracic aortic injury
Blunt trauma
Endovascular prosthesis
Anesthesia
Intensive care
Case report

ABSTRACT

Blunt thoracic aortic injury (BTAI) is a rare but life-threatening consequence of high-energy trauma, most commonly involving the aortic isthmus. Contrast-enhanced computed tomography is essential for rapid diagnosis and therapeutic planning. Thoracic endovascular aortic repair (TEVAR) has become the preferred treatment in hemodynamically stable patients because of its lower perioperative morbidity compared with open surgery. We report 2 cases of traumatic aortic isthmus pseudoaneurysm successfully treated with emergency TEVAR. A 34-year-old male motorcyclist presented with a 31 × 28 mm isthmic pseudoaneurysm associated with bilateral hemothorax and pulmonary contusions. A 17-year-old male sustained a 23 × 38 mm saccular pseudoaneurysm with pneumothorax following a similar mechanism of injury. Both patients underwent urgent TEVAR under general anesthesia using percutaneous femoral access; in the second case, adjunctive left subclavian artery stenting was required due to lesion proximity to the supra-aortic branches. Postoperative intensive care management included close hemodynamic and respiratory monitoring. These cases underscore the importance of rapid imaging, multidisciplinary coordination,

☆ Competing Interests: The authors have declared that no competing interests exist.

☆☆ Acknowledgments: We would like to thank the medical and nursing teams of Mohammed VI University Hospital for their significant involvement in the management of the patients included in our case report. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

* Corresponding author.
E-mail address: taloubi.mohammed.16@gmail.com (M. Taloubi).

https://doi.org/10.1016/j.radcr.2026.06.050

1930-0433/© 2026 The Author(s). Published by Elsevier Inc. on behalf of University of Washington. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/)

and comprehensive perioperative management including anesthetic planning and critical care monitoring in the successful early management of BTAI. In our 2 patients, emergency TEVAR was associated with favorable early clinical outcomes, supporting its feasibility as part of a multidisciplinary treatment strategy in appropriately selected cases.

© 2026 The Author(s). Published by Elsevier Inc. on behalf of University of Washington.
This is an open access article under the CC BY-NC-ND license
(http://creativecommons.org/licenses/by-nc-nd/4.0/)

Introduction

Blunt thoracic aortic injury (BTAI) is a life-threatening condition and the second leading cause of death following nonpenetrating trauma, after intracranial hemorrhage. It most commonly results from high-energy rapid deceleration, which also determines the severity of the lesion [1]. Over 60% of BTAIs occur at the aortic isthmus, where the relatively fixed descending aorta transitions to the more mobile aortic arch [2].

Thoracic endovascular aortic repair (TEVAR) has emerged as the preferred first-line treatment for traumatic isthmic injuries, avoiding thoracotomy, aortic cross-clamping, and cardiopulmonary bypass. Despite its growing use, challenges remain regarding imaging assessment, procedural planning, and preservation of supra-aortic vessels [3].

We present 2 cases of post-traumatic aortic isthmus pseudoaneurysm successfully treated with endovascular repair, highlighting key imaging findings, multidisciplinary decision-making, and technical considerations relevant to trauma and critical care management.

Case I

A 34-year-old male patient was admitted to the emergency department following a high-energy road traffic accident involving a motorcyclist struck by a vehicle, with documented cranial and thoracic impact. On admission, the patient was conscious with a Glasgow Coma Scale score of 15/15. Pupils were equal, round, and reactive to light.

Hemodynamic assessment revealed a blood pressure of 142/78 mmHg and a heart rate of 89 beats per minute. Respiratory evaluation demonstrated a respiratory rate of 18 breaths per minute and an oxygen saturation of 96% on room air.

Physical examination revealed multiple thoracic abrasions associated with tenderness over the right shoulder. Pleuropulmonary auscultation demonstrated decreased breath sounds over the bilateral basal lung fields. The remainder of the systemic examination was unremarkable.

The patient's past medical history was significant for previous surgical treatment of a hepatic cyst and ongoing chronic tobacco use.

As part of the institutional severe trauma protocol, whole-body contrast-enhanced computed tomography, including thoracic CT angiography, was performed within the first hour after admission.

Contrast-enhanced computed tomography angiography of the chest demonstrated an adequate opacification of the thoracic aorta with identification, at the level of the aortic isthmus, of a pseudoaneurysmal outpouching measuring 31 mm in maximal diameter and extending over 28 mm in length, consistent with a Grade III injury. No contrast extravasation was observed during the arterial phase. Associated findings included a moderate bilateral hemothorax, resulting in adjacent passive compressive atelectasis of the underlying pulmonary parenchyma (Fig. 1).

Diagnostic assessment included laboratory investigations, which demonstrated a stable hemoglobin level of 12.3 g/dL. Renal function and coagulation parameters were within normal limits, with a serum urea of 0.45 g/L, serum creatinine of 7.64 mg/L, and a prothrombin time of 100%. High-sensitivity troponin level was 32.6 ng/L, within the normal reference range. Electrocardiography showed normal sinus rhythm without evidence of ischemic changes or conduction abnormalities.

Based on multidisciplinary assessment, TEVAR was performed within 24 hours of diagnosis.

Strict anti-impulse therapy was maintained before intervention with target systolic blood pressure below 120 mmHg and heart rate below 90 beats/min whenever clinically tolerated.

The procedure was performed under general anesthesia in a hybrid operating room. After systemic heparinization, percutaneous right common femoral artery access was obtained under ultrasound guidance. A pigtail catheter was positioned in the ascending aorta for angiographic control.

Based on preoperative computed tomography measurements, TEVAR was performed by advancing a 30 × 30 × 100 mm thoracic stent-graft prosthesis over a stiff guidewire and accurately positioning it across the lesion. Controlled hypotension was induced during deployment to prevent device migration. Completion angiography demonstrated complete exclusion of the pseudoaneurysm without endoleak and preserved supra-aortic branch perfusion (Fig. 2). The femoral access site was closed percutaneously, and the patient was transferred to the intensive care unit in stable condition.

The patient remained in the intensive care unit for 3 days and was subsequently transferred to the vascular surgery department. Total hospital stay was 10 days. Follow-up thoracoabdominal CT angiography performed 2 weeks after discharge confirmed correct stent-graft positioning across the aortic isthmus, complete exclusion of the pseudoaneurysm, and no evidence of endoleak (Fig. 3). Subsequent follow-up was conducted in the vascular surgery outpatient clinic.

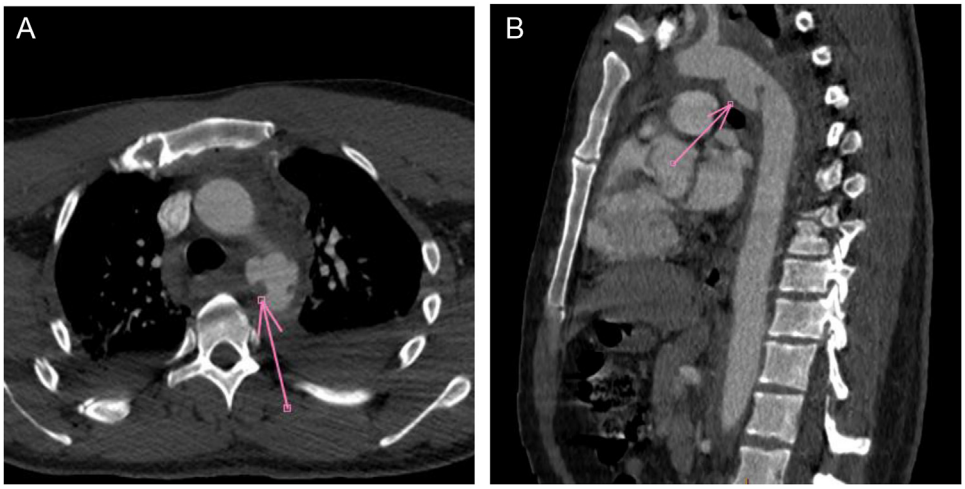


Fig. 1 – Patient 1 – Post-traumatic aortic isthmus pseudoaneurysm on contrast-enhanced CT imaging: (A) axial view and (B) sagittal reconstruction. The arrow indicates the saccular pseudoaneurysm arising from the aortic isthmus.

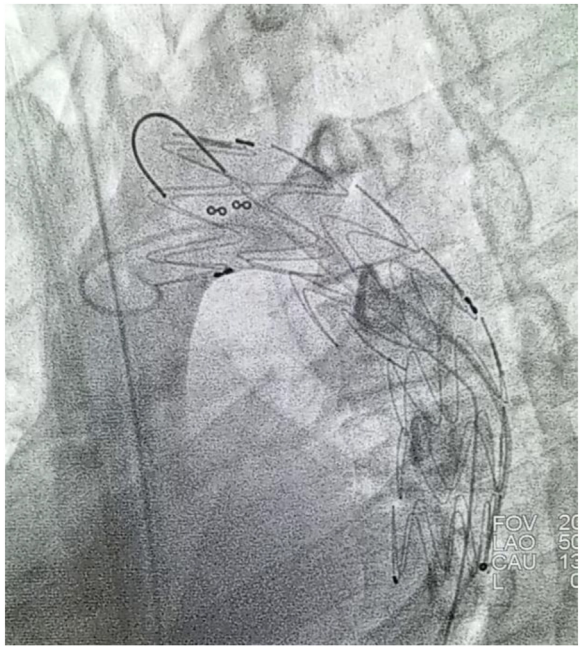


Fig. 2 – Patient 1 – Intraoperative fluoroscopic image during TEVAR showing stent-graft deployment at the aortic isthmus.

Case II

A 17-year-old male patient was admitted to the emergency department following a high-energy road traffic collision involving a motorcyclist struck by a vehicle, resulting in documented cranial and thoracic impact.

On initial assessment, the patient was confused, with a Glasgow Coma Scale score of 13/15. Pupils were equal, round, and reactive to light. Hemodynamic parameters were stable, with a blood pressure of 128/65 mmHg and a heart rate of 90 beats per minute. Respiratory evaluation demonstrated a respiratory rate of 17 breaths per minute and peripheral oxygen saturation of 98% on room air.

Physical examination revealed multiple facial abrasions and a visible deformity overlying the left clavicular region. The chest wall appeared symmetrical with preserved respiratory excursion and bilaterally audible breath sounds on pleuropulmonary auscultation. The remainder of the systemic examination was unremarkable. The patient had no prior medical or surgical history.

According to the institutional severe trauma protocol, whole-body contrast-enhanced computed tomography, including thoracic CT angiography, was performed within the first hour after admission.

Contrast-enhanced computed tomography angiography demonstrated adequate opacification of the thoracic aorta, identifying at the level of the aortic isthmus a saccular contrast-filled outpouching consistent with a traumatic pseudoaneurysm, measuring 23 mm in maximal transverse diameter and extending longitudinally over 38 mm, without evidence of active contrast extravasation during the arterial phase. Associated thoracic injuries included a moderate right-sided pneumothorax and a thin left-sided hemothorax. Additionally, bilateral pulmonary parenchymal ground-glass opacities, more prominent within the left lower lobe, were observed, consistent with traumatic alveolar hemorrhage in the context of blunt thoracic injury (Fig. 4).

Laboratory investigations demonstrated a stable hemoglobin level of 12.6 g/dL. Renal and coagulation parameters were within normal limits, including a serum urea level of 0.30 g/L, serum creatinine of 9.09 mg/L, and a prothrombin time of 77%. Cardiac biomarker analysis revealed a markedly elevated troponin level of 1323 ng/L, with subsequent declining values on serial measurements. Transthoracic echocardiography was normal with no evidence of structural cardiac abnormalities or ventricular dysfunction. In the context of blunt thoracic trauma, the transient troponin elevation was considered compatible with a mild blunt car-

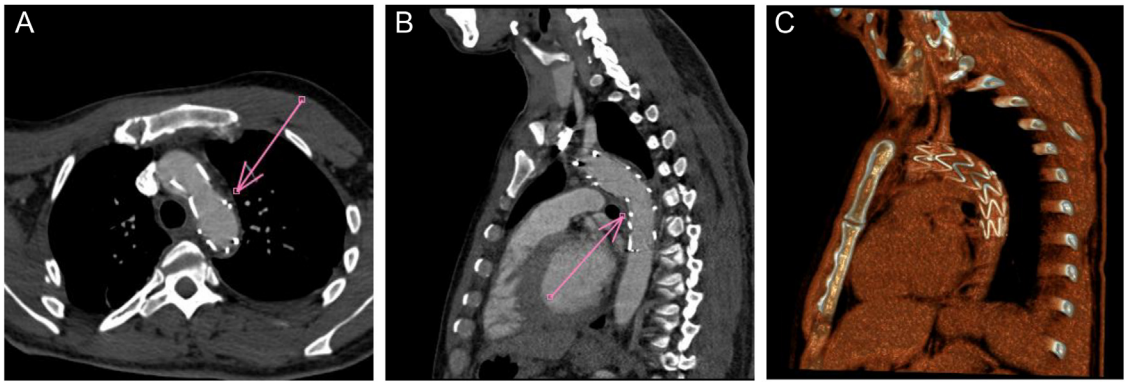


Fig. 3 – Patient 1 – Postoperative contrast-enhanced chest CT following thoracic endovascular aortic repair (TEVAR). (A) Axial image; (B) Sagittal reconstruction confirming appropriate positioning and alignment of the endograft along the descending thoracic aorta (arrow) and (C) Three-dimensional volume-rendered reconstruction.

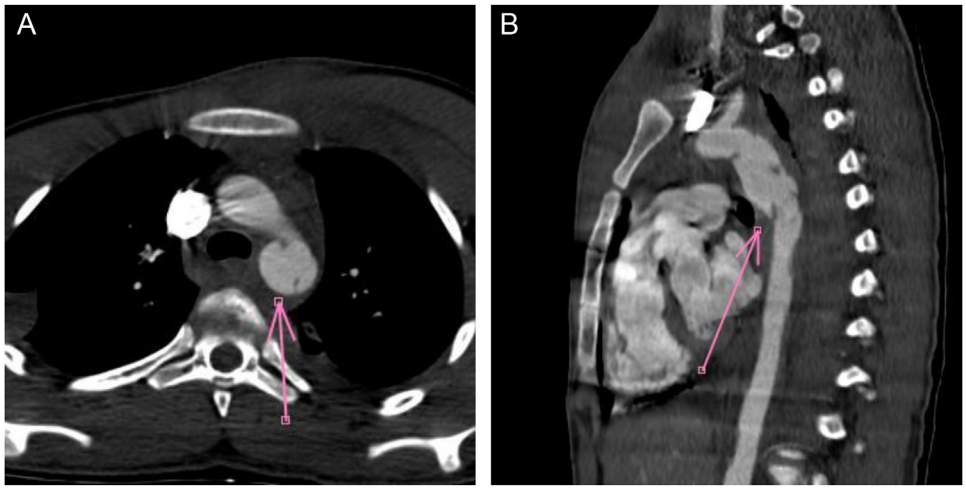


Fig. 4 – Patient 2 – Post-traumatic pseudoaneurysm of the aortic isthmus on contrast-enhanced CT. (A) Axial CT image demonstrating a saccular pseudoaneurysm arising from the aortic isthmus (arrow) and (B) Sagittal reconstructed image confirming the location and longitudinal extent of the aortic isthmus pseudoaneurysm (arrow).

diac injury. Continuous ECG and hemodynamic monitoring were maintained throughout the perioperative period.

Electrocardiography demonstrated normal sinus rhythm without evidence of ischemic abnormalities or conduction disturbances.

Strict anti-impulse therapy was maintained before intervention with target systolic blood pressure below 120 mmHg and heart rate below 90 beats/min whenever clinically tolerated.

Following multidisciplinary evaluation involving trauma surgeons, vascular surgeons, anesthesiologists, and interventional radiologists, TEVAR was performed within 48 hours of diagnosis under general anesthesia in a hybrid operating room for exclusion of the traumatic aortic isthmus pseudoaneurysm. Intraoperative systemic heparinization was administered according to institutional vascular surgery protocol. Percutaneous femoral arterial access was obtained, and a 24 × 24 × 160 mm Ankura thoracic stent-graft was advanced and deployed under fluoroscopic guidance across the injured aortic segment. Due to the proximity of the lesion to the supra-aortic vessels, an 8 × 39 mm self-expanding stent was implanted in the left subclavian artery via humeral access. This strategy was selected to preserve left upper limb perfusion and vertebrobasilar circulation while ensuring an adequate proximal landing zone for the thoracic endograft. Completion angiography confirmed successful exclusion of the pseudoaneurysm without evidence of endoleak (Fig. 5). Postprocedural antithrombotic management consisted of single antiplatelet therapy with aspirin, selected in view of the associated traumatic injuries and bleeding risk.

The postoperative course was favorable, with stable hemodynamic parameters and optimal blood pressure control. Emergence from anesthesia was smooth and uneventful. After confirmation of adequate spontaneous ventilation and neurological responsiveness, tracheal extubation was successfully performed in the operating room. The patient was subse-

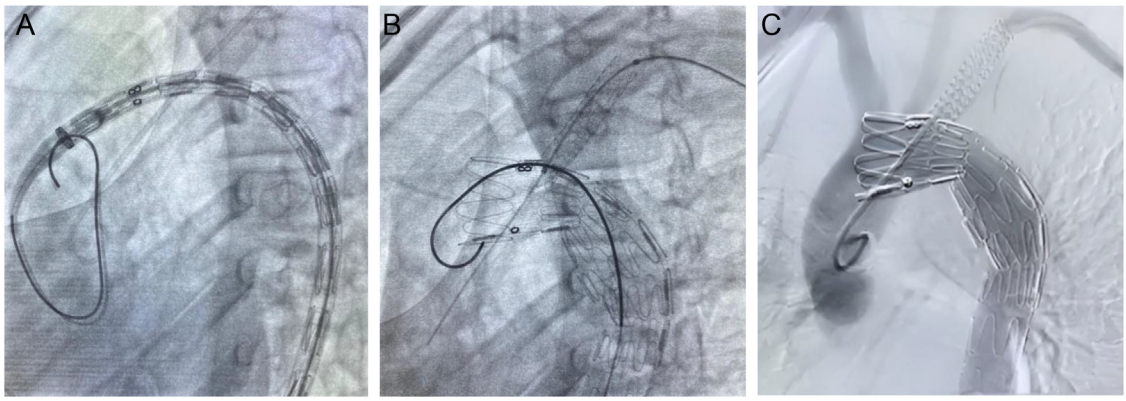


Fig. 5 – Patient 2 – Endovascular treatment of a post-traumatic aortic isthmus pseudoaneurysm. (A) Intraoperative fluoroscopic image showing advancement of the guidewire and delivery system within the thoracic aorta. (B) Deployment of the thoracic stent-graft across the aortic isthmus covering the pseudoaneurysm under fluoroscopic guidance and (C) Completion angiography demonstrating successful exclusion of the pseudoaneurysm following thoracic endograft placement, associated with stent implantation in the left subclavian artery to preserve supra-aortic branch perfusion, without residual endoleak.

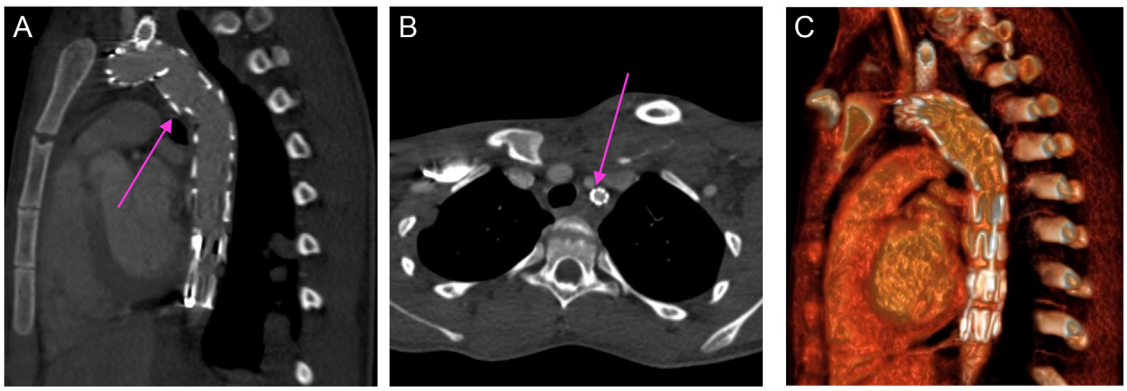


Fig. 6 – Follow-up CT angiography of patient 2 performed 2 weeks after TEVAR and left subclavian artery stenting. (A) Sagittal reconstruction showing satisfactory thoracic stent-graft positioning with complete exclusion of the pseudoaneurysm (arrow). (B) Axial image demonstrating patency of the left subclavian artery stent (arrow) and (c) Three-dimensional volume-rendered reconstruction confirming appropriate positioning of the thoracic endograft and left subclavian artery stent.

quently transferred to the intensive care unit for close postoperative monitoring, including continuous hemodynamic and respiratory surveillance.

The postoperative course remained uneventful. The patient stayed in the intensive care unit for 4 days before transfer to the vascular surgery department. Total hospital stay was 10 days. Follow-up thoracoabdominal CT angiography performed 2 weeks after discharge demonstrated satisfactory thoracic stent-graft positioning, complete exclusion of the pseudoaneurysm without evidence of endoleak, and preserved patency of the left subclavian artery stent (Fig. 6). The patient remained free of upper limb ischemic symptoms and vertebrobasilar manifestations throughout the follow-up period. Ongoing clinical and radiological surveillance was subsequently conducted in the vascular surgery outpatient clinic.

Discussion

Trauma remains a leading cause of mortality among young adults and is most commonly associated with high-energy blunt mechanisms involving rapid deceleration. Motor vehicle collisions, particularly head-on impacts, account for nearly 81% of cases. BTAI results from a combination of shear and stretching forces, abrupt increases in intravascular pressure, and compression of the aorta between the anterior chest wall and the vertebral column [1]. More than 60% of lesions occur at the aortic isthmus, a particularly vulnerable transition zone between the relatively fixed descending thoracic aorta and the mobile aortic arch [2].

The diagnosis of BTAI relies on a high index of clinical suspicion during the initial trauma assessment performed according to Advanced Trauma Life Support principles, with imaging playing a central role. Although chest radiography may demonstrate indirect suggestive findings, computed tomography angiography has become the diagnostic modality of choice owing to its excellent sensitivity and high negative predictive value. Conventional angiography is currently reserved primarily for therapeutic purposes rather than diagnostic evaluation [2].

Based on computed tomography findings, BTAI is classified into 4 grades according to lesion severity. Grade I corresponds to an intimal tear, Grade II corresponds to an intramural hematoma, Grade III corresponds to a pseudoaneurysm reflecting a contained rupture, and Grade IV corresponds to complete aortic rupture, a life-threatening condition requiring immediate intervention [4]. According to this classification, the lesion observed in our cases corresponded to a Grade III injury.

The optimal timing of intervention for Grade III BTAI remains a matter of clinical judgment and should be individualized according to the patient's overall condition and injury characteristics. Although both of our patients were hemodynamically stable and showed no active contrast extravasation on CT angiography, early TEVAR was selected following multidisciplinary discussion involving intensivists, anesthesiologists, vascular surgeons, and radiologists. This decision was supported by the presence of traumatic pseudoaneurysms at the aortic isthmus, the high-energy deceleration mechanism, associated thoracic injuries, and the perceived risk of delayed rupture. Current recommendations support endovascular repair for selected Grade III injuries while emphasizing individualized management based on anatomical, clinical, and multidisciplinary considerations [5,6].

TEVAR, initially validated for the treatment of thoracic aneurysmal disease, has progressively expanded its indications and now plays a major role in the management of other thoracic aortic pathologies, including traumatic aortic isthmus pseudoaneurysm and other blunt thoracic aortic injuries [7,8].

Despite the limited availability of randomized high-level evidence, TEVAR has progressively emerged as the preferred therapeutic strategy for traumatic aortic injuries. Compared with open surgical repair, which requires thoracotomy, aortic cross-clamping, and complex cardiopulmonary support, TEVAR offers reduced operative complexity and lower perioperative morbidity [9]. Polytrauma patients frequently poorly tolerate open surgery because of associated injuries, hemorrhagic risk, and positioning constraints. Furthermore, the increasing availability of endovascular expertise and dedicated equipment has made TEVAR a more accessible and practical option in emergency trauma settings [3].

Although TEVAR may be performed under local anesthesia in elective settings, emergency management in polytrauma patients often involves agitation, lack of cooperation, and multiple concomitant injuries, rendering local anesthesia suboptimal. Current evidence therefore supports the use of general anesthesia in traumatic aortic injury [10]. In our 2 reported cases, the procedures were performed under general anesthesia, with controlled apnea and transient induced hypotension during stent-graft deployment to optimize prosthesis positioning and procedural accuracy.

TEVAR combined with carotid–subclavian bypass or subclavian transposition is widely considered the gold standard for left subclavian artery (LSA) revascularization, providing low morbidity and excellent long-term patency. Revascularization may be performed concomitantly with TEVAR or in a staged manner depending on upper limb perfusion adequacy. The decision to adopt a monitoring strategy rather than immediate revascularization highlights the importance of individualized clinical judgment [11]. In Case 1, preservation of left upper limb perfusion was achieved without the need for additional supra-aortic intervention. Conversely, in our second case, concomitant LSA stenting was successfully performed during TEVAR for the first time at our center, demonstrating the feasibility and potential advantages of integrating endovascular LSA revascularization during aortic stent-graft deployment, particularly in complex anatomical situations where LSA coverage is unavoidable.

The use of thoracic endografts in traumatic aortic injury nevertheless remains technically challenging. Currently available devices are primarily designed for aneurysmal disease and may be suboptimally adapted to the smaller diameters and increased angulation typically observed in younger trauma patients, thereby increasing the risk of endoleak or stent-graft collapse. Accurate device sizing is further complicated by hypovolemia-induced vasospasm and dynamic variations in aortic diameter. However, ongoing technological innovations in endograft design are expected to progressively address these limitations [3].

This report has several limitations that should be acknowledged. First, it describes only 2 patients, limiting the generalizability of the observations. Second, the follow-up period remains relatively short and does not allow assessment of long-term graft durability, late endoleaks, or other delayed complications. Consequently, our findings should be interpreted as illustrative of the feasibility of a multidisciplinary endovascular approach rather than evidence supporting broad clinical recommendations.

Conclusion

Our 2 cases highlight the rarity of traumatic aortic isthmus injuries and underscore the critical importance of rapid diagnosis and timely management. Favorable early outcomes were achieved through a multidisciplinary approach involving intensivists, anesthesiologists, vascular surgeons, and radiologists. In both patients, emergency TEVAR was successfully performed and was associated with satisfactory early clinical outcomes, including in a complex anatomical setting. While these observations support the feasibility of TEVAR in selected patients, larger studies with longer follow-up are needed to further evaluate its safety, effectiveness, and long-term outcomes in the management of traumatic aortic injuries.

Ethical approval

Ethical approval was not required for publication of this case report in accordance with our institutional policy.

Consent to publish

Written informed consent for publication of clinical data and accompanying images was obtained from the adult patient in Case 1 and from the parents/legal guardians of the minor patient in Case 2. Copies of the written consent forms are available for review by the Editor-in-Chief of this journal upon request.

Registration of research studies

This article is a case report and does not constitute an interventional or observational research study involving human participants. Therefore, registration of a research study was not required.

Authors contribution

Mohammed Taloubi: Corresponding author, study concept, Data collection, data analysis, writing review & editing; **Imane Zahrou:** contributor; **Saley Maman Rabilou:** contributor; **Mohammed Chafi:** contributor; **Lamya Ait Sayad:** supervision and data validation; **Ghizlane El Aidouni:** supervision and data validation; **Houssam Bkiyar:** supervision and data validation.

Patient consent

Written informed consent was obtained from the patient for publication of this case report and accompanying images.

REFERENCES

[1] Mouawad NJ, Paulisin J, Hofmeister S, Thomas MB. Blunt thoracic aortic injury: concepts and management. J Cardiothorac Surg 2020;15(1):62. doi:10.1186/s13019-020-01101-6.
[2] Akhmerov A, DuBose J, Azizzadeh A. Blunt thoracic aortic injury: current therapies, outcomes, and challenges. Ann Vasc Dis 2019;12(1):1–5. doi:10.3400/avd.ra.18-00139.
[3] Patelis N, Katsargyris A, Klonaris C. Endovascular repair of traumatic isthmic ruptures: special concerns. Front Surg 2017;4:32. doi:10.3389/fsurg.2017.00032.
[4] Czerny M, Grabenwöger M, Berger T, Aboyans V, Corte AD, et al., Authors/Task Force Members EACTS/STS guidelines for diagnosing and treating acute and chronic syndromes of the aortic organ. Ann Thorac Surg 2024;118(1):5–115. doi:10.1016/j.athoracsur.2024.01.021.
[5] Lee WA, Matsumura JS, Mitchell RS, Farber MA, Greenberg RK, Azizzadeh A, et al. Endovascular repair of traumatic thoracic aortic injury: clinical practice guidelines of the society for vascular surgery. J Vasc Surg 2011;53(1):187–92. doi:10.1016/j.jvs.2010.08.027.
[6] Mouawad NJ, Paulisin J, Hofmeister S, Thomas MB. Blunt thoracic aortic injury: concepts and management. J Cardiothorac Surg 2020;15(1):62. doi:10.1186/s13019-020-01101-6.
[7] Canaud L, Alric P, Branchereau P, Ané CM, Berthet JP. Lessons learned from midterm follow-up of endovascular repair for traumatic rupture of the aortic isthmus. J Vasc Surg 2008;47(4):733–8. doi:10.1016/j.jvs.2007.12.008.
[8] Estrera AL, Gochnour DC, Azizzadeh A, Miller CC, Coogan S, Ouw KC, et al. Progress in the treatment of blunt thoracic aortic injury: 12-year single-institution experience. Ann Thorac Surg 2010;90(1):64–71. doi:10.1016/j.athoracsur.2010.03.053.
[9] Brinster DR. Endovascular repair of blunt thoracic aortic injuries. Semin Thorac Cardiovasc Surg. 2009;21(4):393–8. doi:10.1053/j.semtcvs.2009.11.003.
[10] Lee WA, Matsumura JS, Mitchell RS, Farber MA, Greenberg RK, Azizzadeh A, et al. Endovascular repair of traumatic thoracic aortic injury: clinical practice guidelines of the society for vascular surgery. J Vasc Surg 2011;53(1):187–92. doi:10.1016/j.jvs.2010.08.027.
[11] Zhang Z, Sun G, Ye J, Liu B, Wang Y, Li Y, et al. Comparing endovascular techniques for left subclavian artery revascularization during zone 2 thoracic endovascular aortic repair for type B acute aortic syndromes: a retrospective cohort study. Front Cardiovasc Med 2025;12:1566798. doi:10.3389/fcvm.2025.1566798.